Sharp pointwise lower bounds for Lebesgue functions

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Abstract

For each n, let λ_n be the Lebesgue function for polynomial interpolation at an arbitrary set of n distinct nodes in $[-1,1]$. We prove that there are a fixed point $x \in (-1,1)$ and a constant C such that $\lambda_n(x) > (2/\pi)\log n - C$ for infinitely many n. We also prove that $\limsup_{n\to\infty} \lambda_n(x)/\log n \geq 2/\pi$ for almost every $x \in (-1,1)$. The first conclusion answers the bounded-loss question in the interpretation that the constant may depend on the array and the point; the second answers the almost-everywhere question in Erdős Problem 1132. The first proof combines Tao's local potential estimates with a local Riesz differentiation formula, an energy estimate for nodal derivative jumps, and a second moment argument. The second proof uses positive Cauchy transforms and harmonic measure and is independent of Tao's local Bernstein theory.

1 Introduction

For each $n \geq 1$, let

$$-1 \leq x_{1,n} < \cdots < x_{n,n} \leq 1.$$

Define the node polynomial, the cardinal polynomials, and the Lebesgue function by

$$P_n(x) = \prod_{k=1}^{n} (x - x_{k,n}),$$

$$\ell_{k,n}(x) = \frac{P_n(x)}{(x - x_{k,n})P_n'(x_{k,n})}, \qquad \lambda_n(x) = \sum_{k=1}^{n} |\ell_{k,n}(x)|.$$

The cardinal polynomials are interpreted by continuity at the nodes. The rows of the array need not be nested.

Theorem 1. *For every triangular array of distinct nodes as above, the following statements hold.*

(i) There exist $x \in (-1,1)$ and a finite constant C such that

$$\lambda_n(x) > \frac{2}{\pi}\log n - C \tag{1.1}$$

for infinitely many n.

(ii) For Lebesgue almost every $x \in (-1,1)$,

$$\limsup_{n\to\infty} \frac{\lambda_n(x)}{\log n} \geq \frac{2}{\pi}. \tag{1.2}$$

Erdős posed both questions for triangular arrays in [3, p. 68], and they are recorded as Problem 1132 in [1]. The constant in (1.1) is independent of n; it may depend on the array and the point x.

The global lower bound for Lebesgue constants has a classical history. Faber proved the order $\log n$, Bernstein obtained the asymptotically sharp leading constant $2/\pi$, Erdős and Turán improved the error to $O(\log\log n)$, Erdős to $O(1)$, and Vértesi determined the optimal asymptotics through the constant term. See [5, Section 1.3] and [2] for references and further historical discussion.

For triangular arrays, Bernstein's result already implies that the set of points x satisfying

$$\limsup_{n\to\infty} \frac{\lambda_n(x)}{\log n} \geq \frac{2}{\pi}$$

is everywhere dense; Erdős explicitly asked whether this holds for almost every x [3, p. 68]. Erdős and Vértesi later proved the weaker almost-everywhere conclusion $\limsup_{n\to\infty} \lambda_n(x) = \infty$ for arbitrary triangular arrays [4]. Thus Theorem 1(ii) gives the sharp logarithmic rate almost everywhere.

Tao [5, Theorem 1.10(i)] proved that every fixed nondegenerate interval $I \subset [-1, 1]$ satisfies

$$\sup_{x\in I} \lambda_n(x) \geq \frac{2}{\pi} \log n - O_I(1).$$

His Corollary 1.11 and Remark 1.12 give, for every prescribed $\omega(n) \to \infty$, a dense, indeed comeager, set of points x for which

$$\lambda_n(x) \geq \frac{2}{\pi} \log n - \omega(n)$$

infinitely often. Theorem 1(i) gives a fixed additive constant at a fixed point. In its proof, Baire category is used only to produce an interval with a uniform upper bound; the recurrence on a set of positive measure comes from the energy estimate in Section 3.

Throughout the paper, all logarithms are natural and $|E|$ denotes Lebesgue measure. We write $c_0 = 2/\pi$, and suppress the row index when discussing a single row. Constants in local estimates may depend on the fixed intervals and test functions, but not on n or on interpolation data bounded by one. The notation $J \Subset U$ means that J is a compact subset of the open set U.

2 A local differentiation formula

We use the following local form of Riesz's differentiation formula and its near-equality consequence.

Lemma 2. *Suppose that F is holomorphic in a neighborhood of the square $\{|\Re z|, |\Im z| \leq T\}$, is real on its real diameter, and satisfies, throughout this square,*

$$|F(u + iv)| \leq e^{|v|}.$$

For sufficiently large T,

$$|F'(0)| \leq 1 + O(T^{-1}). \tag{2.1}$$

If $F'(0) \geq 1 - \delta$, where $\delta \geq 0$, then

$$F(\pi/2) \geq 1 - \frac{\pi^2}{4}\delta - O(T^{-1}). \tag{2.2}$$

The constants are absolute.

Proof. Choose $R = m\pi$ with $T/2 \le R < T$, and integrate

$$\frac{F(z)}{z^2 \cos z}$$

around the square of radius R. On its vertical sides, $|\cos z| = \cosh(\Im z)$; on its horizontal sides, $|\cos z| \ge \sinh R$. The contour integral, divided by $2\pi i$, is consequently $O(R^{-1})$. Taking residues at zero and at $t_j = (j + \frac{1}{2})\pi$ gives

$$F'(0) = \sum_{|t_j|<R} \frac{(-1)^j F(t_j)}{t_j^2} + O(T^{-1}). \tag{2.3}$$

Since $|F(t_j)| \le 1$ and

$$\sum_{j\in\mathbb{Z}} \frac{1}{t_j^2} = 1,$$

equation (2.1) follows. Isolating the term $t_0 = \pi/2$ in (2.3) gives

$$1 - \delta \le \frac{4}{\pi^2} F(\pi/2) + 1 - \frac{4}{\pi^2} + O(T^{-1}),$$

which proves (2.2). □

We apply this formula to interpolants with bounded nodal data.

Lemma 3. *Let $I \Subset (-1, 1)$ be a fixed nondegenerate closed interval. Suppose that, for all sufficiently large n,*

$$\lambda_n(x) \le \Lambda_n := c_0 \log n + C \qquad (x \in I), \tag{2.4}$$

where C is fixed. There are functions ρ_n on $\operatorname{int} I$ whose restrictions to each fixed compact interval are bounded above and below by positive constants and have uniformly bounded Lipschitz norms. They have the following properties on every fixed interval $J \Subset \operatorname{int} I$.

For every real polynomial

$$q(x) = \sum_{k=1}^{n} a_k \ell_{k,n}(x), \qquad |a_k| \le 1,$$

uniformly for $x \in J$,

$$|q'(x)| \le \pi n \rho_n(x)(\Lambda_n + o(1)), \qquad |q''(x)| \le C_J n^2 \log n. \tag{2.5}$$

Moreover, for every fixed D, if

$$\frac{|q'(x)|}{\pi n \rho_n(x)} \ge c_0 \log n - D, \tag{2.6}$$

then there is $y = x + O_J(n^{-1})$ such that

$$\lambda_n(y) \ge c_0 \log n - D', \tag{2.7}$$

where D' depends on I, J, C, D, but not on n, x, q. The point y lies in any fixed compact interior interval containing J in its interior, for all sufficiently large n.

Finally, if h_n have support in a fixed compact subset of $\operatorname{int} I$ and have uniformly bounded Lipschitz norms, then

$$\frac{1}{n} \sum_k h_n(x_{k,n}) = \int h_n(x)\rho_n(x)\, dx + O(n^{-1/10}). \tag{2.8}$$

Proof. Fix one compact interval $I' \Subset \operatorname{int} I$ with $J \Subset \operatorname{int} I'$, also containing, when (2.8) is being proved, the fixed compact set containing the supports of the functions h_n. Assumption (2.4) implies the hypothesis $\sup_I \lambda_n \ll n^{O(1)}$ of Tao [5, Theorem 4.1, equation (4.1)]. Put

$$S_n = \sum_k \frac{1}{|P_n'(x_{k,n})|}, \qquad \alpha_n = \frac{1}{n}\log S_n, \qquad U_n(z) = \frac{1}{n}\log\frac{1}{|P_n(z)|}.$$

Tao's equation (4.11) identifies this α_n, while Theorem 4.1(ii),(iv), equations (4.6), (4.7), and (4.9), give on I' the asserted upper and lower bounds and Lipschitz bound for ρ_n, together with

$$U_n(x+i\eta) = \alpha_n - \pi\eta\rho_n(x) + O\Big(\frac{\log n}{n} + \eta^3\Big) \tag{2.9}$$

for $\eta \geq \log n/n$, uniformly on fixed compact interior intervals. It also gives, for every subinterval K of such an interval,

$$\frac{1}{n}\#\{k : x_{k,n} \in K\} = \int_K \rho_n(x)\,dx + O\bigg(\frac{|K|}{A} + \frac{A^2\log n}{n}\bigg) \qquad (A \geq 1). \tag{2.10}$$

The function ρ_n here is Tao's density defined from the original interval I by his equation (4.15), so it does not depend on the auxiliary choice of I'. Taking $A = n^{1/10}$ and integrating by parts proves (2.8).

For complex z with $\Re z \in I'$ and $\Im z = \eta \geq \log n/n$, the interpolation formula and (2.9) imply

$$|q(z)| \leq \frac{S_n|P_n(z)|}{\eta} \leq \frac{1}{\eta}\exp\Big(\pi n\eta\rho_n(\Re z) + O(\log n + n\eta^3)\Big). \tag{2.11}$$

Fix $x \in J$, and set

$$L = n^{-1/5}, \qquad H = n^{-2/5}, \qquad \omega = \pi n\rho_n(x)(1 + D_0 n^{-1/10}),$$

where $D_0 > 0$ is any fixed constant. On the upper edge of the rectangle

$$|\Re z - x| \leq L, \qquad 0 \leq \Im z \leq H,$$

the Lipschitz bound for ρ_n and (2.11) give

$$|q(z)| \leq \Lambda_n e^{\omega H}$$

for all sufficiently large n: the allowance in ωH is of order $n^{1/2}$, while the density variation contributes $O(n^{2/5})$ and the remaining terms contribute $O(\log n)$.

On the lower edge, $|q| \leq \lambda_n \leq \Lambda_n$. Expanding q in Chebyshev polynomials on I also gives the bound $|q(z)| \leq \Lambda_n \exp(O_I(n))$ on the vertical sides. Apply the local Phragmén–Lindelöf estimate [5, Lemma 2.5] to

$$\Lambda_n^{-1} q(z) e^{i\omega(z-x)}.$$

In the notation of that lemma, the upper and lower edge bound is 1, the vertical-edge factor is $M = \exp(O_I(n))$, the height is H, and points with $|u| \leq L/2$ have distance at least $L/2$ from the vertical sides. Thus its error is $\exp(O(ne^{-cL/H}))=1 + O(n^{-10})$. For $|u| \leq L/2$ and $0 \leq v \leq H$, it yields

$$|q(x+u+iv)| \leq \Lambda_n(1 + n^{-10})e^{\omega|v|}. \tag{2.12}$$

Reflection gives the estimate for negative v.

After the change of variable $t = \omega(z - x)$, the function

$$F(t) = \frac{q(x + t/\omega)}{\Lambda_n(1 + n^{-10})}$$

satisfies Lemma 2 on a square of radius $T \asymp_J n^{3/5}$. Equation (2.1), and the fact that $\Lambda_n n^{-1/10} = o(1)$, give the first estimate in (2.5). Cauchy's estimate on a disk of radius comparable to n^{-1} in (2.12) gives the second estimate.

Under (2.6), replacing q by $-q$ if necessary, we have

$$F'(0) \geq 1 - \frac{C_1}{\Lambda_n}$$

with C_1 independent of n, x, q. Equation (2.2) now shows that

$$|q(x + \pi/(2\omega))| \geq \Lambda_n - C_2.$$

Since $|q(y)| \leq \lambda_n(y)$ for real y, this proves (2.7). □

3 Derivative jumps and a Riesz energy

At each internal interpolation node, write

$$J_k = \lambda_n'(x_{k,n}+) - \lambda_n'(x_{k,n}-).$$

At the first and last nodes, use the exterior polynomial pieces. Set

$$\nu_k = \frac{1}{S_n |P_n'(x_{k,n})|}.$$

The absolute values of all cardinal polynomials other than $\ell_{k,n}$ have a cusp at $x_{k,n}$, whereas $\ell_{k,n}$ is positive near that node. Consequently

$$J_k = 2 \sum_{j \neq k} |\ell_{j,n}'(x_{k,n})| = 2 \sum_{j \neq k} \frac{\nu_j}{\nu_k |x_{k,n} - x_{j,n}|}. \tag{3.1}$$

Lemma 4. *Under the hypotheses of Lemma 3, choose a nonzero nonnegative smooth function f compactly supported in* $\operatorname{int} I$. *Then*

$$\sum_k \frac{f(x_{k,n})^2}{n\rho_n(x_{k,n})} \frac{J_k}{2\pi n \rho_n(x_{k,n})} \geq c_0 \log n \int f(x)^2 \, dx - O(1). \tag{3.2}$$

Terms with $f(x_{k,n}) = 0$ are taken to be zero; ρ_n is only needed on a fixed interior neighborhood of the support of f.

Proof. Put $a_k = f(x_{k,n})/\rho_n(x_{k,n})$, with $a_k = 0$ outside the support of f, and define

$$\mu_n = \frac{1}{n} \sum_k a_k \delta_{x_{k,n}}, \qquad K_\varepsilon(t) = \frac{1}{\sqrt{t^2 + \varepsilon^2}}.$$

The kernel K_ε is positive definite, since

$$K_\varepsilon(t) = \frac{1}{\sqrt{\pi}} \int_0^\infty s^{-1/2} e^{-s\varepsilon^2} e^{-st^2} \, ds$$

is a positive mixture of Gaussian kernels. Write

$$\mathcal{E}_\varepsilon(\sigma,\tau) = \iint K_\varepsilon(x-y)\, d\sigma(x)\, d\tau(y)$$

for its bilinear energy. Positivity applied to $\mu_n - f(x)\,dx$ gives

$$\mathcal{E}_\varepsilon(\mu_n,\mu_n) \geq 2\mathcal{E}_\varepsilon(\mu_n, f\,dx) - \mathcal{E}_\varepsilon(f\,dx, f\,dx). \tag{3.3}$$

Take $\varepsilon = 1/n$. Subtracting $f(x)$ in the integral over $|t| \leq 1$, and using the Lipschitz bound for f, gives the uniform estimate

$$(K_{1/n} * f)(x) = 2\log n\, f(x) + O_f(1). \tag{3.4}$$

The mass of μ_n is bounded. Also, by (2.8),

$$\int f\, d\mu_n = \frac{1}{n}\sum_k \frac{f(x_{k,n})^2}{\rho_n(x_{k,n})} = \int f^2 + O(n^{-1/10}). \tag{3.5}$$

Equations (3.3)–(3.5) therefore yield

$$\mathcal{E}_{1/n}(\mu_n,\mu_n) \geq 2\log n \int f^2 - O(1). \tag{3.6}$$

The diagonal part of this energy is $n^{-1}\sum_k a_k^2 = O(1)$. Since $K_{1/n}(t) \leq 1/|t|$ for $t \neq 0$,

$$\frac{1}{n^2}\sum_{k\neq j} \frac{a_k a_j}{|x_{k,n}-x_{j,n}|} \geq 2\log n \int f^2 - O(1). \tag{3.7}$$

Finally,

$$a_k^2 \frac{\nu_j}{\nu_k} + a_j^2 \frac{\nu_k}{\nu_j} \geq 2a_k a_j.$$

By (3.1), the left side of (3.2) is

$$\frac{1}{\pi n^2}\sum_{k\neq j} \frac{a_k^2 \nu_j}{\nu_k |x_{k,n}-x_{j,n}|}.$$

Pairing the terms indexed by (k,j) and (j,k), and then applying (3.7), bounds this expression below by

$$\frac{1}{\pi n^2}\sum_{k\neq j} \frac{a_k a_j}{|x_{k,n}-x_{j,n}|} \geq \frac{2}{\pi}\log n \int f^2 - O(1).$$

This proves (3.2). □

4 A fixed point with bounded additive loss

Proposition 5. *Assume the local upper bound* (2.4) *on a fixed nondegenerate interval* $I \Subset (-1,1)$*. There are a constant* D *and a measurable set* $G \subset \operatorname{int} I$*, with* $|G| > 0$*, such that every* $x \in G$ *satisfies*

$$\lambda_n(x) \geq c_0 \log n - D$$

for infinitely many n*.*

Proof. Choose compact intervals $J_0 \Subset J_1 \Subset \operatorname{int} I$, and choose f as in Lemma 4 with $\operatorname{supp} f \subset J_0$. All constants below are independent of n: after I, C, J_0, J_1, f are fixed, we choose D, then $\kappa, D_1, d, \kappa_1, d_1$, and finally b, D_2. Let

$$w_k = \frac{f(x_{k,n})^2}{n\rho_n(x_{k,n})}, \qquad T_k = \frac{J_k}{2\pi n\rho_n(x_{k,n})}, \qquad W = \int f^2 > 0.$$

By (2.8),

$$\sum_k w_k = W + O(n^{-1/10}), \qquad 0 \le w_k \le C_f/n. \tag{4.1}$$

On each interval between successive nodes, λ_n equals an interpolant with nodal data in $\{-1, 1\}$. Both one-sided derivatives at a node therefore satisfy (2.5), so

$$0 \le T_k \le \Lambda_n + o(1) \tag{4.2}$$

uniformly on the support of f. Lemma 4, (4.1), and (4.2) show that

$$\sum_k w_k(\Lambda_n + 1 - T_k) = O(1), \tag{4.3}$$

with nonnegative summands for all sufficiently large n. On a node with $T_k < c_0 \log n - D$, the factor in (4.3) is greater than $C + 1 + D$. Hence, by choosing D sufficiently large, such nodes carry at most $W/2$ of the weights. Since (4.1) has total weight $W + o(1)$, for all sufficiently large n the nodes in the support of f satisfying

$$T_k \ge c_0 \log n - D \tag{4.4}$$

carry at least $W/4$ of the weights. In particular, there are at least κn such nodes, with $\kappa > 0$ independent of n. Discarding the first and last nodes does not change this conclusion after reducing κ.

At a node satisfying (4.4), at least one adjacent polynomial piece q has

$$|q'(x_{k,n})| \ge \pi n\rho_n(x_{k,n})(c_0 \log n - D).$$

Lemma 3 gives a point

$$y_k = x_{k,n} + O(n^{-1}), \qquad \lambda_n(y_k) \ge c_0 \log n - D_1. \tag{4.5}$$

For a node satisfying (4.4), take an adjacent polynomial piece with the large derivative above. Its values at the two endpoints of the corresponding gap are both 1. Rolle's theorem and (2.5) imply that this gap has length at least d/n for some fixed $d > 0$ when the gap is contained in J_1. If it is not contained in J_1, its length is bounded below by $\operatorname{dist}(J_0, \mathbb{R} \setminus J_1) > 0$.

An interval of length less than d/n contains at most two such nodes, so every interval of length $O(n^{-1})$ contains only a bounded number of the nodes in (4.4). By (4.5), the same holds, with multiplicities, for the points y_k. Hence one can select at least $\kappa_1 n$ of them, separated by d_1/n, for fixed $\kappa_1, d_1 > 0$.

Equation (2.5), applied to every polynomial piece, also shows that λ_n is $C_I n \log n$-Lipschitz on a fixed compact interior interval containing these points. Choose a fixed sufficiently small $b > 0$, and let A_n be the union of the intervals of radius

$$r_n = \frac{b}{n \log n}$$

centered at the selected points. For all sufficiently large n, these intervals are disjoint, lie in a fixed compact interior interval, and satisfy

$$\lambda_n(x) \geq c_0 \log n - D_2 \quad (x \in A_n), \qquad \frac{c}{\log n} \leq |A_n| \leq \frac{C_2}{\log n}. \tag{4.6}$$

The separation of the centers in the m-th row implies, for every interval B,

$$|A_m \cap B| \leq C_3 \left(\frac{|B|}{\log m} + \frac{1}{m \log m} \right).$$

Since A_n has at most n components, for $m > n$,

$$|A_n \cap A_m| \leq C_3 \left(\frac{|A_n|}{\log m} + \frac{n}{m \log m} \right). \tag{4.7}$$

Put $B_j = A_{2^j}$. Equations (4.6) and (4.7) give

$$|B_j| \asymp j^{-1}, \qquad |B_j \cap B_k| \leq C_4 \left(\frac{1}{jk} + \frac{2^{j-k}}{k} \right) \quad (j < k). \tag{4.8}$$

For every fixed j_0, the sum of the measures of B_j, $j_0 \leq j \leq N$, grows at least as a positive constant times $\log N$. By (4.8), the integral of the square of the sum of their indicators is $O((\log N)^2)$. Cauchy–Schwarz consequently gives a positive lower bound, independent of j_0, for

$$\left| \bigcup_{j \geq j_0} B_j \right|.$$

Continuity of measure along these decreasing tail unions shows that

$$|\limsup_{j \to \infty} B_j| > 0. \tag{4.9}$$

Together with (4.6), this proves the proposition. $\square$

Proof of Theorem 1(i). Suppose the assertion were false. Then

$$\lambda_n(x) - c_0 \log n \longrightarrow -\infty \qquad \text{for every } x \in (-1, 1). \tag{4.10}$$

Fix a nondegenerate compact interval $K_0 \subset (-1, 1)$. The closed sets

$$F_M = \{x \in K_0 : \lambda_n(x) \leq c_0 \log n + M \text{ for every } n \geq 2\}, \qquad M = 1, 2, \ldots,$$

cover K_0, by (4.10). Baire's theorem supplies a nondegenerate closed subinterval $I \subset K_0$ on which (2.4) holds with a fixed C. Proposition 5 contradicts (4.10). Increasing the final additive constant by one gives the strict inequality in the statement. $\square$

5 Positive Cauchy transforms

Suppress the row index and put

$$S = \sum_k \frac{1}{|P'(x_k)|}, \qquad \nu_k = \frac{1}{S|P'(x_k)|}, \qquad \epsilon_k = \operatorname{sgn} P'(x_k).$$

Define

$$m(z) = \sum_k \frac{\nu_k}{z - x_k}, \qquad g(z) = \sum_k \frac{\epsilon_k \nu_k}{z - x_k} = \frac{1}{SP(z)}. \tag{5.1}$$

The last identity is the partial fraction formula for $1/P$, whose expansion at infinity gives

$$\sum_k \epsilon_k \nu_k x_k^j = 0 \qquad (0 \le j \le n-2). \tag{5.2}$$

For $n \ge 2$, both $m + g$ and $m - g$ are Cauchy transforms of positive probability measures: their masses at x_k are $(1 \pm \epsilon_k)\nu_k$.

For real x away from the nodes, write

$$H(x) = \sum_k \frac{\nu_k}{|x - x_k|}, \qquad \gamma_s(x) = -\Im m(x + is), \qquad p_s(t) = \frac{s}{\pi(t^2 + s^2)}.$$

Then

$$\lambda_n(x) = \frac{H(x)}{|g(x)|}, \qquad H(x) = \frac{2}{\pi} \int_0^\infty \gamma_s(x) \, \frac{ds}{s}, \tag{5.3}$$

and the Poisson semigroup gives

$$\gamma_s = p_{s-h} * \gamma_h \qquad (s > h). \tag{5.4}$$

Lemma 6. *For every row with $n \ge 2$, every $0 < h \le 1$, and every $x \in [-1, 1]$,*

$$\frac{|g(x + ih)|}{\gamma_h(x)} \le \frac{4 + h^2}{h^2 \sinh((n-1)h/2)}. \tag{5.5}$$

Consequently, if $0 < a < 1$ and $h = n^{-a}$, then

$$\sup_{x \in [-1,1]} \frac{|g(x + ih)|}{\gamma_h(x)} \longrightarrow 0. \tag{5.6}$$

Proof. Set $z = x + ih$ and $p = T_{n-1}$, the Chebyshev polynomial of the first kind. The polynomial $(p(t) - p(z))/(t - z)$ has degree at most $n - 2$, so (5.2) gives

$$p(z)g(z) = \sum_k \frac{\epsilon_k \nu_k p(x_k)}{z - x_k}, \qquad |p(z)g(z)| \le \frac{1}{h},$$

using $|p(x_k)| \le 1$ and $\sum_k \nu_k = 1$. Write $z = \cos(u + iv)$, with $u, v \in \mathbb{R}$. Then $h \le \sinh |v|$, so $|v| \ge \operatorname{arsinh} h \ge h/2$. The identity $T_{n-1}(\cos w) = \cos((n-1)w)$ yields

$$|p(z)|^2 = \cos^2((n-1)u) + \sinh^2((n-1)v) \ge \sinh^2((n-1)h/2).$$

Finally, since the nodes and x lie in $[-1, 1]$,

$$\frac{h}{4 + h^2} \le \gamma_h(x) = \sum_k \frac{\nu_k h}{(x - x_k)^2 + h^2} \le \frac{1}{h}.$$

Combining these bounds proves (5.5). For $h = n^{-a}$, its right-hand side tends to zero: the hyperbolic sine grows exponentially in n^{1-a}, while h^{-2} grows polynomially. This proves (5.6). □

Lemma 7. *Let F be the Cauchy transform of a positive probability measure with finite support. For every bounded real interval S_0, every $x \in \mathbb{R}$, and every $h > 0$,*

$$\int_{\mathbb{R}} p_h(x-y)\,\mathbf{1}_{\{F(y)\in S_0\}}\,dy = \frac{1}{\pi}\int_{S_0} \frac{-\Im F(x+ih)}{(t-\Re F(x+ih))^2 + (\Im F(x+ih))^2}\,dt. \tag{5.7}$$

Proof. Compose harmonic measure of S_0 in the lower half-plane with F. The resulting bounded harmonic function on the upper half-plane has the indicated boundary values almost everywhere. The exceptional boundary points are poles of F and preimages of the endpoints of S_0, a finite set. Its Poisson representation is (5.7). □

6 The almost-everywhere bound

Proof of Theorem 1(ii). Suppose that the assertion fails. By a countable-union argument, there are a measurable set $E \subset (-1,1)$ of positive measure, a constant $0 < c < c_0$, and an integer N such that

$$\lambda_n(y) \le c\log n \qquad (y \in E,\ n \ge N). \tag{6.1}$$

Choose constants

$$K > 1, \qquad 0 < b < a < 1, \qquad \frac{2b}{\pi c K} > 1,$$

which is possible because $c < 2/\pi$. Set

$$h = n^{-a}, \qquad \eta = n^{-b}, \qquad H^\eta(x) = \frac{2}{\pi}\int_\eta^1 \gamma_s(x)\,\frac{ds}{s}.$$

By (5.3), $H^\eta \le H$. For $s > 0$ and real x, y, t, the triangle inequality gives

$$|y + is - t| \le |x + is - t| + |x - y| \le \left(1 + \frac{|x-y|}{s}\right)|x + is - t|.$$

Taking reciprocal squares and summing against the positive masses yields $\gamma_s(x) \le (1 + |x - y|/s)^2\gamma_s(y)$. Integrating over $\eta \le s \le 1$, we obtain

$$H^\eta(x) \le \left(1 + \frac{|x-y|}{\eta}\right)^2 H^\eta(y). \tag{6.2}$$

Fix $x \in E$, and write

$$u = \Re m(x+ih), \qquad \gamma = \gamma_h(x), \qquad S_x = [u - K\gamma, u + K\gamma].$$

By (5.6) and (5.7), each of the two events

$$m(y) + g(y) \in S_x, \qquad m(y) - g(y) \in S_x$$

has probability

$$p_K + o(1), \qquad p_K = \frac{2}{\pi}\arctan K > \frac{1}{2},$$

under the Poisson probability $p_h(x-y)\,dy$. The error is uniform in $x \in E$. Their intersection therefore has probability at least

$$q_K + o(1), \qquad q_K = 2p_K - 1 > 0. \tag{6.3}$$

Choose R so that the Poisson mass of $|y-x| > Rh$ is less than $q_K/8$, and let

$$E_n = \{x \in E : (p_h * \mathbf{1}_{\mathbb{R}\setminus E})(x) < q_K/8\}.$$

The approximate identity theorem gives $|E \setminus E_n| \to 0$. For all sufficiently large n, (6.3) implies that, for every $x \in E_n$, there exists $y \in E$ with $|y-x| \le Rh$ and both events true. We may take y away from the nodes, and then

$$|g(y)| \le K\gamma_h(x).$$

Using (5.3), (6.1), and (6.2), we obtain

$$H^\eta(x) \le (1+Rh/\eta)^2 H^\eta(y) \le cK(1+Rh/\eta)^2(\log n)\gamma_h(x) \qquad (x \in E_n). \tag{6.4}$$

Let T_n be convolution with the symmetric probability kernel

$$k_n(t) = \frac{1}{b\log n}\int_\eta^1 p_{s-h}(t)\,\frac{ds}{s}.$$

Equation (5.4) shows that

$$H^\eta = \frac{2b}{\pi}(\log n)T_n\gamma_h.$$

Thus (6.4) becomes

$$\frac{T_n\gamma_h(x)}{\gamma_h(x)} \le \frac{1}{A_n} \quad (x \in E_n), \qquad A_n = \frac{2b}{\pi cK}(1+Rh/\eta)^{-2} \longrightarrow A > 1. \tag{6.5}$$

Integrate (6.5) over E_n, discard the contribution from $y \notin E_n$, and symmetrize. Positivity of γ_h, symmetry of k_n, and $(r+r^{-1})/2 \ge 1$ give

$$\frac{|E_n|}{A_n} \ge \iint_{E_n\times E_n} k_n(x-y)\frac{\gamma_h(y)}{\gamma_h(x)}\,dy\,dx \ge \iint_{E_n\times E_n} k_n(x-y)\,dy\,dx. \tag{6.6}$$

The kernels k_n form an approximate identity. In fact, for every fixed $\delta > 0$,

$$\int_{|t|>\delta} k_n(t)\,dt \le \frac{C}{b\log n}\int_\eta^1 \min(1, s/\delta)\,\frac{ds}{s} \longrightarrow 0.$$

It follows that

$$\iint_{E\times E} k_n(x-y)\,dy\,dx \longrightarrow |E|.$$

Replacing E by E_n changes this integral by at most $2|E\setminus E_n|$. Taking limits in (6.6) therefore yields

$$\frac{|E|}{A} \ge |E|,$$

contrary to $|E| > 0$ and $A > 1$. This proves the assertion. $\square$

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GPT-6 Astra was used to generate the mathematical proofs and draft the manuscript. GPT-5.6 Sol and Claude Opus 5 were used for editorial review of the exposition. The author reviewed the final manuscript and takes full responsibility for its content.

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